

Differential Equations Lesson 03

Question 1:

Solve the differential equation $\frac{dy}{dx} = \frac{4x^3}{y^2}$

a) Convert $\frac{dy}{dx} = \frac{4x^3}{y^2}$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}?$

Question 2:

Solve the differential equation $\frac{dy}{dx} = \frac{4 - x^2}{5y^3}$

a) Convert $\frac{dy}{dx} = \frac{4 - x^2}{5y^3}$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}?$

Question 3:

Solve the differential equation $(7 + 3x)\frac{dy}{dx} = 4y$

a) Convert $(7 + 3x)\frac{dy}{dx} = 4y$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}$

Question 4:

Solve the differential equation $(5 + 3x^2) \cdot y' = 4xy$

a) Convert $(5 + 3x^2) \cdot y' = 4xy$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}$

Question 5:

Solve the differential equation $\sqrt{3x^3 + 8} \cdot y' = 4x^2 y$

a) Convert $\sqrt{3x^3 + 8} \cdot y' = 4x^2 y$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

Question 6:

Solve the differential equation $\sqrt{x^2 - 12} \cdot y' = 4xy$

a) Convert $\sqrt{x^2 - 12} \cdot y' = 4xy$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}$

Question 7:

Solve the differential equation $y \ln x^3 - xy' = 0$.

a) Convert $y \ln x^3 - xy' = 0$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}$

Question 8:

Solve the differential equation $y \ln x^{1/3} - 4xy' = 0$.

a) Convert $y \ln x^{1/3} - 4xy' = 0$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}$

Question 9:

Find the general solution of the differential equation $\sqrt{x} + y \cdot y' = 0$.

- a) Convert $\sqrt{x} + y \cdot y' = 0$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$
- b) General solution of the differential equation is $\underline{\hspace{1cm}}$
- c) Find the particular solution that satisfies initial condition $y(0) = 4$.

Question 10:

Find the general solution of the differential equation $y(4 + x^2) - x(5 - y^3) \cdot y' = 0$.

a) Convert $y(4 + x^2) - x(5 - y^3) \cdot y' = 0$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}}?$

c) Find the particular solution that satisfies initial condition $y(0) = -3$.

Question 11:

Find the general solution of the differential equation $\frac{dy}{dx} = \frac{4 \sin 5x}{y^{1/2}}$

a) Convert $\frac{dy}{dx} = \frac{4 \sin 5x}{y^{1/2}}$ to differential form: $\underline{\hspace{1cm}} \cdot dy = \underline{\hspace{1cm}} \cdot dx$

b) General solution of the differential equation is $\underline{\hspace{1cm}} ? \hspace{1cm}$

c) Find the particular solution that satisfies initial condition $y(\pi/2) = 1$.

Note: When $x = \pi/2$, $y = 1$

Question 12:

Find the general solution of the differential equation $\frac{dy}{dx} = \frac{3 \tan x}{y^2}$

a) Convert $\frac{dy}{dx} = \frac{3 \tan x}{y^2}$ to differential form: $\underline{\quad ? \quad} \cdot dy = \underline{\quad ? \quad} \cdot dx$

b) General solution of the differential equation is $\underline{\quad ? \quad}$

c) Find the particular solution that satisfies initial condition $y(\pi/2) = 1$.

Note: When $x = \pi/2$, $y = 1$

